

01 TEAM DETAILS

UdyamPulse

An explainable MSME Financial Health Card built for IDBI Innovate 2026.

Team name	Looper
Team leader name	Bhunesh Bansal
Problem Statement	Problem Statement 3 — Financial Health Score

02 BRIEF ABOUT THE IDEA

Consented alternate data, turned into a decision a bank can defend.

01 What it is.

An explainable MSME Financial Health Card that scores businesses from consented alternate data.

03 Inputs.

GST filings, UPI transaction trails, Account Aggregator bank statements, EPFO records.

02 Who it's for.

New-to-Credit (NTC) and New-to-Bank (NTB) MSMEs that traditional bureau-based underwriting rejects outright.

04 Outputs.

A 0–100 score, an A–E grade, an eligible credit limit, and a plain-language improvement plan.

Explainability and auditability, by default.

DIFFERENTIATION

Beyond a number

Most alternate-data scoring tools return a score. UdyamPulse returns a decision a bank can defend to a regulator and a borrower can act on.

APPROACH

Two engines, one verdict

Independent, cross-checkable scoring methods run side by side and show explicitly what bureau-only underwriting would have decided vs. alternate data.

USP

The regulatory gap

Of 127 AI-using regulated entities surveyed, only 15 use SHAP/LIME explainability and only 18 keep audit logs (RBI). UdyamPulse ships both by default.

What the health card actually does.

5-Pillar Health Score

Liquidity, Discipline, Momentum, Leverage, Digital Footprint → 0–100 score, A–E grade, eligible limit.

Explainable ML + Shapley

A trained model with exact per-feature Shapley attribution behind every decision — no black box.

Traditional vs. Alternate Verdict

Side-by-side comparison showing the rejected-to-approved reversal for thin-file businesses.

Improvement Plan + Audit Log

A concrete action plan for borrowers, and an append-only audit trail for every decision made.

From consented data to an auditable decision.

01

Data intake

AA statements, GST, UPI, EPFO



02

Dual scoring

Rule engine + ML model, in parallel



03

Verdict & explain

Traditional vs. alternate, Shapley reasons



04

Memo & audit

Underwriter memo, decision logged

The rejected → approved moment.

The NTC hero (Shree Ganesh Textiles) has zero bureau history — declined instantly by a traditional process, approved on alternate data by UdyamPulse.

TRADITIONAL (BUREAU-ONLY)

Rejected

No credit bureau history on file (New-to-Credit / New-to-Bank).

UDYAMPULSE (ALTERNATE DATA)

Approved

Grade A · Score 86/100 · Eligible limit ₹27,00,000.

One service. Two engines. A full audit trail.

Single FastAPI Service

One process serves the API and the frontend — one Dockerfile, one deploy target.

Rule + ML Dual Engine

Transparent 5-pillar rules alongside a trained model with exact Shapley attribution.

Memo Layer

Deterministic by default; optional AWS Bedrock Runtime provider is wired for Stage 2.

Audit & Compliance Log

Pseudonymous HMAC subjects, restart-safe SHA-256 chaining, role-gated access.

A reproducible, inspectable open stack.

Backend	Python, FastAPI, Pydantic, pytest
ML layer	Calibrated monotonic XGBoost champion with native exact TreeSHAP; calibrated logistic fallback
Frontend	Vanilla HTML / CSS / JS, no build step
Deployment & CI	Docker, Render web service, render.yaml Blueprint, GitHub Actions
Stage 2 targets	IDBI sandbox credentials, dated MSME repayment labels, SSO/KMS, durable shared audit storage

09 ESTIMATED IMPLEMENTATION COST (optional)

Near-zero to prototype. Scoped jointly for Stage 2.

The prototype runs entirely on an open-source stack and free-tier hosting. Stage 2 production cost scales with transaction volume and is best scoped jointly with IDBI during the sandbox mentorship phase, once real data volume is known.

The live decision stage.

Captured from the verified submission build: a calm NTC reversal stage with a top review index and half-page slide-over packets for evidence, governance, proof, and source map detail.

UdyamPulse

MSME credit review, without hiding the audit trail.

Live API / xgboost pd proxy v1 model

Public synthetic cohort with sandbox feed contracts

Decision

Evidence

Governance

Proof

Sources

Borrower file

Shree Ganesh Textiles (New-to-Credit)

Shree Ganesh Textiles (New-to-Credit) Textiles / Surat A grade / ₹27,00,000	Patel Wholesale Traders Wholesale / Ahmedabad B grade / ₹32,80,000	City Corner Retail Retail / Nagpur D grade / ₹1,80,000	Sunrise Auto Parts Auto Components / Pune C grade / ₹7,75,000	Himalayan H Handicrafts / D B grade /
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UP-ntc-hero

Textiles / Surat

New-to-Credit / New-to-Bank

Bureau rejection reversed

Low risk

A

86/100

Eligible limit

Live first viewport: one clear NTC rejection reversal, top review index, borrower strip, and stamped approval card.

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City Corner Retail

Decision packet

Decision slide-over: pillar ledger, reason-code journal, memo, and borrower improvement note.

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Governance packet

Governance slide-over: audit count, validation metrics, pilot KPIs, fairness slices, controls, and source signals.

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Proof packet

Proof tab: truth boundary, judging rubric, competitor gap map, runbook, and API catalog.

Verified, not just claimed.

50/50

Automated tests passing

0.750

Untouched holdout AUC

0.499

Holdout Gini

0.012

Calibration error

Measured on a locked 4,500-row public-proxy holdout: AUC 0.7497 (95% bootstrap CI 0.7314-0.7678), KS 0.4225, Brier 0.1415, and PSI 0.0003. This is cross-sectional evidence, not IDBI calibration or true out-of-time proof.

Stage 2 is plug-in ready.

Phase 1

Real data integration

Connect authenticated IDBI sandbox AA/GST/UPI/EPFO feeds to /sandbox/score and /sandbox/recalibration/report; recalibrate on real distributions.

Phase 2

IDBI-calibrated champion

Retrain the shipped XGBoost/logistic champion-challenger pipeline on dated sandbox outcomes and validate a true OOT window.

Phase 3

Fairness & monitoring

Run the implemented fairness gap, PSI, and reason-code stability checks on production-scale data.

Phase 4

Bedrock-powered memos

Enable the optional AWS Bedrock provider in the pilot environment, with deterministic fallback.

13 LINKS

Where to find everything.

→ GitHub Public Repository <https://github.com/bansalbhunesh/id>

→ Recorded Walkthrough github.com/bansalbhunesh/id/blob/main/docs/demo.webm

→ Final Product Link <https://id-ysm9.onrender.com>